
Topic: Additional *PROBLEMS* on Waves

Problem 1:

Given the wave equation:

$$y(x, t) = (3.00 \text{ mm}) \sin[(4.00 \text{ m}^{-1})x - (7.00 \text{ s}^{-1})t],$$

[a] Find speed of the wave.

Solution

comparing it to the general wave equation:

$$y(x, t) = y_m \sin(kx - \omega t).$$

From the given equation, we identify:

$$\begin{aligned} k &= 4.00 \text{ m}^{-1}, \\ \omega &= 7.00 \text{ rad/s}. \end{aligned}$$

The speed of the wave is given by:

$$v = \frac{\omega}{k}.$$

Substituting the given values:

$$\begin{aligned} v &= \frac{7.00 \text{ rad/s}}{4.00 \text{ m}^{-1}} \\ &= 1.75 \text{ m/s}. \end{aligned}$$

Thus, the speed of the wave is 1.75 m/s.

Problem 2:

A rope, under a tension of 200 N and fixed at both ends, oscillates in a second-harmonic standing wave pattern. The displacement of the rope is given by:

$$y = (0.10 \text{ m}) \sin\left(\frac{\pi x}{2}\right) \sin(12\pi t), \quad (1)$$

where $x = 0$ at one end of the rope, x is in meters, and t is in seconds. Find:

- (a) The length of the rope.
- (b) The speed of the waves on the rope.
- (c) The mass of the rope.
- (d) The period of oscillation if the rope oscillates in a third-harmonic standing wave pattern.

Solution

(a) Length of the Rope

Comparing the given function with the general standing wave equation, we obtain:

$$k = \frac{\pi}{2}, \quad \omega = 12\pi \text{ rad/s}.$$

Since $k = \frac{2\pi}{\lambda}$, we solve for λ :

$$\begin{aligned} \frac{2\pi}{\lambda} &= \frac{\pi}{2}, \\ \lambda &= 4.0 \text{ m}. \end{aligned}$$

For the second harmonic, $L = \lambda$, so:

$$L = 4.0 \text{ m}. \quad (2)$$

(b) Speed of the Wave

Since $\omega = 2\pi f$, we solve for f :

$$\begin{aligned} 2\pi f &= 12\pi, \\ f &= 6.0 \text{ Hz}. \end{aligned}$$

Using $v = f\lambda$, we find:

$$\begin{aligned} v &= (6.0 \text{ Hz})(4.0 \text{ m}), \\ v &= 24 \text{ m/s}. \end{aligned}$$

(c) Mass of the Rope

Using the wave speed equation:

$$v = \sqrt{\frac{T}{\mu}}, \quad (3)$$

where $T = 200$ N and $\mu = \frac{m}{L}$. Solving for m :

$$\begin{aligned} 24 &= \sqrt{\frac{200}{m/4.0}}, \\ 24^2 &= \frac{200 \times 4.0}{m}, \\ m &= 1.4 \text{ kg.} \end{aligned}$$

(d) Period in the Third Harmonic

For the third harmonic:

$$f_3 = \frac{3v}{2L} = \frac{3(24)}{2(4.0)} = 9.0 \text{ Hz.} \quad (4)$$

The period is:

$$T = \frac{1}{f_3} = \frac{1}{9.0} = 0.11 \text{ s.} \quad (5)$$

Problem 3:

Two identical traveling waves, moving in the same direction, are out of phase by $\frac{\pi}{2}$ rad. What is the amplitude of the resultant wave in terms of the common amplitude y_m of the two combining waves?

Solution

The displacement of the string is given by:

$$y = y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \phi). \quad (6)$$

Using the trigonometric identity:

$$\sin \alpha + \sin \beta = 2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right), \quad (7)$$

we rewrite the equation as:

$$y = 2y_m \cos \left(\frac{\phi}{2} \right) \sin \left(kx - \omega t + \frac{\phi}{2} \right). \quad (8)$$

Since the two waves are out of phase by $\phi = \frac{\pi}{2}$, we substitute:

$$\begin{aligned} A &= 2y_m \cos \left(\frac{\pi}{4} \right), \\ &= 2y_m \times \frac{\sqrt{2}}{2}, \\ &= \sqrt{2}y_m, \\ &\approx 1.41y_m. \end{aligned}$$

Thus, the amplitude of the resultant wave is $1.41y_m$.

Problem 4:

A string under tension T_i oscillates in the third harmonic at frequency f_3 , and the waves on the string have wavelength λ_3 . If the tension is increased to $T_f = 4T_i$ and the string is again made to oscillate in the third harmonic, determine:

- (a) The new frequency of oscillation in terms of f_3 .
- (b) The new wavelength of the waves in terms of λ_3 .

Solution

Using the equation for resonant frequencies:

$$f_n = \frac{nv}{2L} = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad (9)$$

for the third harmonic:

$$f_3 = \frac{3}{2L} \sqrt{\frac{T_i}{\mu}}. \quad (10)$$

(a) New Frequency

When the tension is increased to $T_f = 4T_i$, the new frequency is:

$$\begin{aligned} f'_3 &= \frac{3}{2L} \sqrt{\frac{T_f}{\mu}} \\ &= \frac{3}{2L} \sqrt{\frac{4T_i}{\mu}} \\ &= \frac{3}{2L} \times 2 \sqrt{\frac{T_i}{\mu}} \\ &= 2f_3. \end{aligned}$$

Thus, the new frequency is $2f_3$.

(b) New Wavelength

The wave speed is given by:

$$v = \sqrt{\frac{T}{\mu}}, \quad (11)$$

so the new wave speed is:

$$v' = \sqrt{\frac{T_f}{\mu}} = \sqrt{\frac{4T_i}{\mu}} = 2v.$$

Since wavelength is given by:

$$\lambda_3 = \frac{v}{f_3}, \quad (12)$$

we find the new wavelength:

$$\lambda'_3 = \frac{v'}{f'_3} = \frac{2v}{2f_3} = \lambda_3.$$

Thus, the wavelength remains unchanged, $\lambda'_3 = \lambda_3$.

Problem 5:

A wire that is 10.0 m long, has a mass of 100 g, and is stretched under a tension of 250 N. Determine:

- (a) The lowest frequency of standing waves.
- (b) The second lowest frequency.
- (c) The third lowest frequency.

Solution

A string clamped at both ends can oscillate in standing wave patterns. The possible wavelengths are given by:

$$\lambda_n = \frac{2L}{n}, \quad (13)$$

where L is the length of the wire and n is an integer.

The corresponding frequencies are:

$$f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}, \quad (14)$$

where v is the wave speed.

The wave speed is given by:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{TL}{M}}, \quad (15)$$

where T is the tension in the wire, μ is the linear mass density, and M is the total mass of the wire. Since $\mu = M/L$, we obtain:

$$f_n = \frac{n}{2} \sqrt{\frac{T}{ML}}. \quad (16)$$

Substituting the given values:

$$\begin{aligned} f_n &= \frac{n}{2} \sqrt{\frac{250}{(0.100)(10.0)}} \\ &= \frac{n}{2} \times \sqrt{250} \\ &= \frac{n}{2} \times 15.81 \\ &= n \times 7.91 \text{ Hz.} \end{aligned}$$

(a) Lowest Frequency

For $n = 1$:

$$f_1 = 1 \times 7.91 = 7.91 \text{ Hz.} \quad (17)$$

(b) Second Lowest Frequency

For $n = 2$:

$$f_2 = 2 \times 7.91 = 15.8 \text{ Hz.} \quad (18)$$

(c) Third Lowest Frequency

For $n = 3$:

$$f_3 = 3 \times 7.91 = 23.7 \text{ Hz.} \quad (19)$$

Thus, the three lowest frequencies are:

$$\begin{aligned} f_1 &= 7.91 \text{ Hz,} \\ f_2 &= 15.8 \text{ Hz,} \\ f_3 &= 23.7 \text{ Hz.} \end{aligned}$$